Поперечные колебания стержня

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In[ ]:=
```
ClearAll["Global`*"]
```

Краевые условия задачи

$$X(0) = X'(0) = 0, \quad X''(l) = 0$$

In[57]:=
```
X1 = Sin[Pi / λ * (x - x0)]
```

Out[57]=

$$\mathrm{Sin}\left[\frac{\pi (x - x0)}{\lambda}\right]$$

In[58]:=
```
X2 = C[1] * Exp[-Pi * x / λ] + C[2] * Exp[Pi * (x - l) / λ]
```

Out[58]=

$$e^{-\frac{\pi x}{\lambda}} c_1 + e^{\frac{\pi (-l+x)}{\lambda}} c_2$$

Уравнения для краевых условий

In[59]:=
```
eq01 = ((X1 + X2[[1]]) /. x → 0) == 0
```

Out[59]=

$$c_1 - \mathrm{Sin}\left[\frac{\pi\, x0}{\lambda}\right] == 0$$

In[60]:=
```
eq02 = MultiplySides[Simplify[(D[X1 + X2[[1]], x] /. x → 0) == 0], λ, Assumptions→{λ > 0}]
```
Out[60]=

$$c_1 - \mathrm{Cos}\left[\frac{\pi\, x0}{\lambda}\right] == 0$$

In[61]:=
```
eql1 = MultiplySides[Simplify[(D[X1 + X2[[2]], {x, 2}] /. x → l) == 0], λ, Assumptions→{λ > 0}]
```
Out[61]=

$$c_2 - \mathrm{Sin}\left[\frac{\pi\,(l - x0)}{\lambda}\right] == 0$$

In[62]:=
```
eql2 = MultiplySides[Simplify[(D[X1 + X2[[2]], {x, 3}] /. x → l) == 0], λ, Assumptions→{λ > 0}]
```
Out[62]=

$$c_2 - \mathrm{Cos}\left[\frac{\pi\,(l - x0)}{\lambda}\right] == 0$$

Поиск угловой скорости

In[64]:=
```
eq0 = eq01[[1, 2, 2, 1]] == Pi / 4 + Pi * k
```
Out[64]=

$$\frac{\pi\, x0}{\lambda} == \frac{\pi}{4} + k\,\pi$$

In[65]:=
```
eq1 = eql1[[1, 2, 2, 1]] == Pi / 4 + Pi * n
```
Out[65]=

$$\frac{\pi\,(l - x0)}{\lambda} == \frac{\pi}{4} + n\,\pi$$

Ответ

In[77]:=
```
sol = Simplify[Solve[eq0 && eq1, {x0, λ}][[1]] /. n → (m - k)]
```

Out[77]=

$$\left\{x0 \to \frac{l + 4\,k\,l}{2 + 4\,m},\ \lambda \to \frac{2\,l}{1 + 2\,m}\right\}$$

In[82]:=
```
ω = Simplify[1 / a * (Pi / sol[[2, 2]])^2];
Print["ω = ", ω, ",    m = 0, 1, 2..."]
```

$$\omega = \frac{(\pi + 2\,m\,\pi)^2}{4\,a\,l^2},\quad m = 0,\ 1,\ 2...$$